

Once methylated, genes usually stay that way through successive cell divisions in a given individual. At DNA sites where one strand is already methylated, enzymes methylate the correct daughter strand after each round of DNA replication. Methylation patterns are thus passed on to daughter cells, and cells forming specialized tissues keep a chemical record of what occurred during embryonic development. A methylation pattern maintained in this way also accounts for *genomic imprinting* in mammals, where methylation permanently regulates expression of either the maternal or paternal allele of particular genes at the start of development (see Figure 15.17). DNA methylation and histone modification are believed to be coordinated in their regulation.

Epigenetic Inheritance

The chromatin modifications that we just discussed do not change the DNA sequence, yet they still may be passed along to future generations of cells. Inheritance of traits transmitted by mechanisms not involving the nucleotide sequence itself is called *epigenetic inheritance*, the study of which is called **epigenetics**. Whereas mutations in the DNA are permanent changes, modifications to the chromatin can be reversed. For example, DNA methylation patterns are largely erased during gamete formation and reestablished during embryonic development. Furthermore, they are changeable, thus responding more rapidly to environmental conditions.

Research on epigenetics has skyrocketed over the past 20 years. The importance of epigenetic information in regulating gene expression is now widely accepted. One key study by Robert Waterland and Randy Jirtle at Duke University, published in 2003, used a mouse mutant whose genome had been altered so that a gene called *agouti* that determines coat color, normally expressed only briefly during fur formation, was instead expressed throughout development. This overexpression resulted in yellow mice (Figure 18.8a, left mouse) rather than the usual brownish (“*agouti*”) color. Previous work had shown that merely supplementing the diet of pregnant mothers with methyl group–containing compounds (like folic acid) could shift the range of coat colors of the offspring back to normal (see Figure 18.8a, right mouse). Waterland and Jirtle reproduced this result, analyzing the state of methylation of the DNA. They showed that the extent of the color shift correlated with the level of DNA methylation. In other words, feeding methyl groups to the mothers at a key time during gestation led to a change in gene expression in the offspring’s phenotype. Further studies showed that the effects were even observed in the next generation—the “grandpups” of the original female mouse.

A similar epigenetic effect due to changes in methylation occurs in humans as well. Near the end of World War II, during the winter of 1944–45, Dutch railway workers went on strike to try to prevent the Nazis from bringing in more troops. In retaliation, the Nazis blocked all deliveries of food. Over 20,000 Dutch died in the “Dutch Hunger Winter”

▼ **Figure 18.8** Examples of epigenetic inheritance.



(a) Effects of maternal diet on genetically identical mice. These two mice are genetically identical (*agouti* mutants), but their mothers were fed different diets. A maternal diet without compounds that donate methyl groups resulted in the obese, yellow offspring on the left. A diet supplemented with methyl-donating compounds, such as folic acid, led to the speckled brown fur (normal *agouti* phenotype) and normal weight of the mouse on the right.



(b) The Dutch Hunger Winter. During the Nazi blockade of food transport into the Netherlands near the end of World War II, the population was starving, including this mother and her two children. Children of mothers who were pregnant during that time have experienced long-term effects.

(Figure 18.8b). Over time, doctors found that the offspring of women in early pregnancy at that time experienced adverse health effects as adults: higher rates of obesity, high triglyceride and cholesterol levels, type 2 diabetes, and schizophrenia. Furthermore, the affected individuals had a 10% higher mortality rate after the age of 68 than their siblings who were *in utero* when food was readily available. A collaboration between labs in the Netherlands and the United States compared those adults with their siblings and published their findings in 2018. Statistical analysis enabled the researchers to conclude that differences between the siblings in DNA methylation of certain genes caused these long-term adverse medical conditions—examples of epigenetic inheritance.

Epigenetic variations might help explain cases where one identical twin acquires a genetically based disease, such as schizophrenia, but the other does not, despite their identical genomes. Alterations in normal patterns of DNA methylation